

**ILIOFEMORAL RIVER**

**INTRUDERS. REPEATS UNINVITED GUESTS, DO NOT BELONG TO THE IV HEPARIN HUNTING FAMILY.**

**NO INTRUDING**

CONTINUING ESTABLISHED FOR BIVALIRUDIN, BUT BIVALIRUDINS HAVE IGNORED IT.

**DIRECT THROMBIN INHIBITORS — NO ANTITHROMBIN NEEDED.**

NO COACH

**BEAVER DAM**

INTO

GRABS SIRE HIDING INSIDE

**CAN INHIBIT BOTH FREE AND CLOT-BOUND THROMBIN**

**UNIVALENT ONE HAND**

TO WRESTLER THROM-BEAVERS WITH BARE HANDS

**ARGATROBAN**

WEIGHT TAG 507 ARA SMALL MOLECULE

**MW 527**

**CLOCK**  $t_{1/2}$  ~45 min

HEPATIC CLEARANCE

**LIVER**

SAFE IN SEVERE RENAL FAILURE ✓  
- NOT REALLY WHY PATIENT HAS KIDNEY FAILURE

GONE HAND ONLY

GRABS THROM-BEAVERS BY THE ACTIVE SITE

CAUTION IN LIVER DISEASE ⚠

**CONFUSED**

**INR/PT MALFUNCTIONING - warfarin scenes**

**MONITORING**  
aPTT → TARGET 1.5–3× BASELINE (NOT >100 seconds)

**THE INR TRAP**

ARGATROBAN PROLONGS INR → COMPLICATES WARFARIN TRANSITION.  
INR OVERESTIMATES ANTICOAGULATION DURING OVERLAP  
SOLUTION 1: USE FACTOR X LEVELS (not affected by argatroban)  
SOLUTION 2: STOP ARGATROBAN 2–3 HOURS BEFORE CHECKING INR

**INDICATIONS**

- LICENSED FOR HIT
- HIT + PCI
- PREFERRED WHEN HIT + SEVERE RENAL FAILURE

**COMPARISON SCOREBOARD**

|               | ARGATROBAN                   | BIVALIRUDIN                        |
|---------------|------------------------------|------------------------------------|
| Binding       | UNIVALENT (active site only) | DIVALENT (active site + exosite 1) |
| MW            | 527                          | 1,980                              |
| $t_{1/2}$     | ~45 min                      | ~25 min (SHORTEST)                 |
| Clearance     | HEPATIC                      | Peptidases + partial renal         |
| Renal safe?   | ✓ YES                        | ⚠ Partial renal                    |
| Liver safe?   | ⚠ NO                         | ✓ YES                              |
| Monitoring    | aPTT 1.5–3×                  | ACT (cath lab) or aPTT             |
| Prolongs INR? | ⚠ YES → Factor X levels      | MINIMAL                            |
| Main use      | HIT + renal failure          | PCI + HIT procedures               |

**TWO-ARMED**

**BOTH HANDS**

ONE GRABS ACTIVE SITE

GRABS EXOSITE 1

**BIVALIRUDIN**

**MW 1,980**

**CLOCK**  $t_{1/2}$  ~25 min — SHORTEST DTI

SCISSORS CLEARED BY PEPTIDASES + PARTIAL RENAL

**PEPTIDASES**

CRACKED KIDNEY ⚠ PARTIAL RENAL CLEARANCE

**MONITORING**  
AT HIGH DOSES (CATH LAB)  
aPTT AT LOWER DOSES

**ACT**

**INR EFFECT**  
MINIMAL INR PROLONGATION

**INDICATIONS**

- ALTERNATIVE TO HEPARIN IN PCI
- HIT + PCI OR CARDIAC BYPASS
- ICU SETTINGS

## 1. Direct thrombin inhibitors

### WHAT YOU SEE

A 'NO INTRUDING' sign with 'no antithrombin needed' — the DTIs hit thrombin directly, no antithrombin middleman required.

### WHAT IT ENCODES

Direct thrombin inhibitors (DTIs) bind and inactivate thrombin (factor IIa) DIRECTLY, with no need for the antithrombin cofactor that heparins require ('No Intruding' = no middleman). Because they act at the final common enzyme of coagulation, they block fibrinogen→fibrin conversion, thrombin-mediated activation of factors V/VIII/XI/XIII, and thrombin-driven platelet activation. The parenteral DTIs are argatroban and bivalirudin (also desirudin/lepirudin historically); the oral DTI is dabigatran.

### BOARD TRAP

Classic board contrast: heparin/LMWH/fondaparinux work INDIRECTLY by potentiating antithrombin, so they fail in antithrombin deficiency and in HIT (where heparin is the culprit). DTIs work independently of antithrombin, which is exactly why they are the agents of choice in HIT.

## 2. Inhibit free + clot-bound

### WHAT YOU SEE

A central banner 'CAN INHIBIT BOTH FREE AND CLOT-BOUND THROMBIN' — DTIs reach thrombin already trapped in the clot, which the bulky heparin-antithrombin complex cannot.

### WHAT IT ENCODES

DTIs inhibit BOTH circulating (free) thrombin AND thrombin already bound to fibrin within an existing clot. This is a key pharmacologic advantage: the heparin–antithrombin complex is too bulky to reach thrombin trapped in a clot, so heparin only neutralizes free thrombin and leaves clot-bound thrombin to keep propagating the thrombus.

### BOARD TRAP

Board favorite: 'Which anticoagulant inactivates clot-bound thrombin?' → a DTI. Heparin/LMWH inactivate only free (fluid-phase) thrombin, allowing rebound thrombin generation from the clot surface.

## 3. Argatroban univalent

### WHAT YOU SEE

A one-armed wrestler labelled 'UNIVALENT ONE HAND' grabbing thrombin's active site with his single arm — one hand = univalent, binding only the catalytic active site.

### WHAT IT ENCODES

Argatroban is a small synthetic UNIVALENT DTI ('one hand') that binds only the catalytic active site of thrombin. It is a reversible competitive inhibitor. Univalent binding means it does not occupy exosite 1.

### BOARD TRAP

Distinguish from bivalirudin (bivalent, binds active site + exosite 1). Hirudin-derived agents are bivalent; argatroban and dabigatran are univalent active-site binders.

## 4. Argatroban hepatic clearance

### WHAT YOU SEE

A liver organ beside argatroban marked 'HEPATIC clearance' — liver-cleared, so it's the DTI safe to use when the kidneys have failed.

### WHAT IT ENCODES

Argatroban is cleared HEPATICALLY (metabolized by the liver, minimal renal excretion). This makes it the DTI of choice in patients with renal failure/dialysis, because the dose does not need adjustment for kidney function. Standard infusion ~2 µg/kg/min, titrated to aPTT 1.5–3× baseline.

### BOARD TRAP

REDUCE or avoid argatroban in significant hepatic impairment (start ~0.5 µg/kg/min in liver disease) — accumulation causes bleeding. Mirror trap: bivalirudin/dabigatran depend on renal clearance, so the rule is 'bad kidneys → argatroban, bad liver → not argatroban.'

## 5. Argatroban for HIT

### WHAT YOU SEE

The argatroban INDICATIONS placard listing 'HIT' and 'severe renal failure' — drug of choice for HIT, especially when the kidneys are out.

### WHAT IT ENCODES

Argatroban is licensed for treatment and prophylaxis of thrombosis in heparin-induced thrombocytopenia (HIT), and is the preferred DTI when the HIT patient ALSO has renal failure (hepatic clearance is unaffected by kidneys).

### BOARD TRAP

In HIT you must stop ALL heparin (including LMWH and heparin flushes) and start a non-heparin anticoagulant (DTI such as argatroban, or fondaparinux/DOAC per scenario). Do NOT start warfarin alone in acute HIT — give a parenteral agent first and overlap, because warfarin-induced protein C drop can precipitate venous limb gangrene.

## 6. Argatroban short half-life

### WHAT YOU SEE

A clock beside argatroban reading t½ ~45 min — short half-life, so anticoagulation wears off quickly once the drip stops (no antidote needed).

### WHAT IT ENCODES

Argatroban has a short half-life of roughly 45 minutes, so anticoagulation reverses quickly on stopping the infusion. Monitored and titrated by aPTT (target 1.5–3× baseline; ACT used during PCI).

### BOARD TRAP

There is NO specific antidote for argatroban or bivalirudin — management of bleeding relies on stopping the drug and waiting out the short half-life (± supportive measures). Only dabigatran has a reversal agent (idarucizumab).

## 7. Bivalirudin bivalent

### WHAT YOU SEE

A two-armed wrestler labelled 'TWO-ARMED / BOTH HANDS' gripping thrombin's active site AND exosite 1 — two hands = bivalent, binding both sites.

### WHAT IT ENCODES

Bivalirudin is a synthetic BIVALENT DTI ('two hands') — a hirudin analog that simultaneously binds thrombin's catalytic active site AND exosite 1 (the fibrinogen-binding/substrate-recognition site). Its inhibition is potent but transient because thrombin slowly cleaves the bound bivalirudin, restoring some enzyme activity.

### BOARD TRAP

Bivalent vs univalent is testable: bivalirudin and the hirudins are bivalent; argatroban and dabigatran are univalent active-site-only inhibitors.

## 8. Bivalirudin renal clearance

### WHAT YOU SEE

A cracked, broken kidney beside bivalirudin marked 'PARTIAL RENAL clearance' — the cracked kidney warns it accumulates and needs dose reduction in renal failure.

### WHAT IT ENCODES

Bivalirudin is cleared mostly by enzymatic proteolysis (thrombin-mediated cleavage), with a significant PARTIAL RENAL component (~20%). Dose reduction is required in renal impairment, and it accumulates in severe renal failure/dialysis.

### BOARD TRAP

Reverse of argatroban: in a HIT patient with renal failure choose argatroban (hepatic), not bivalirudin. Bivalirudin is favored in the cath lab/PCI setting rather than for renal-failure inpatients.

## 9. Bivalirudin shortest half-life

### WHAT YOU SEE

A clock beside bivalirudin reading ~25 min — labelled 'SHORTEST', ideal quick on/off control for PCI.

### WHAT IT ENCODES

Bivalirudin has the shortest half-life of the DTIs (~25 minutes), driven mainly by enzymatic degradation. This rapid offset is ideal for procedural anticoagulation during PCI, where quick on/off control limits bleeding.

### BOARD TRAP

Short half-life is an advantage, but remember there is still no specific reversal agent — you simply stop the infusion and wait.

## 10. Bivalirudin for PCI

### WHAT YOU SEE

The bivalirudin INDICATIONS placard listing 'PCI' and 'HIT during PCI procedures' — the cath-lab DTI of choice.

### WHAT IT ENCODES

Bivalirudin is the DTI of choice for percutaneous coronary intervention (PCI), including PCI in patients with HIT or at high bleeding risk, given via IV during the procedure (ACT-guided). It is used as an alternative to heparin ± GP IIb/IIIa inhibitors.

### BOARD TRAP

Board association: 'HIT patient who needs cardiac catheterization/PCI' → bivalirudin. 'HIT patient with renal failure on the ward' → argatroban.

## 11. Comparison scoreboard

### WHAT YOU SEE

A central 'COMPARISON SCOREBOARD' table pitting argatroban vs bivalirudin head-to-head (binding, MW, clearance,  $t_{1/2}$ , INR effect, use).

### WHAT IT ENCODES

Scoreboard summary — Argatroban: univalent, MW ~527, hepatic clearance,  $t_{1/2}$  ~45 min, the bigger INR offender. Bivalirudin: bivalent, MW ~1980, partial renal/enzymatic clearance,  $t_{1/2}$  ~25 min (shortest), used in PCI. Both are parenteral, both monitored by aPTT (ACT in cath lab), both lack a reversal agent.

### BOARD TRAP

Don't confuse with dabigatran (oral DTI): dabigatran is renally cleared, dosed orally, and IS reversible with idarucizumab — the only DTI with a specific antidote.

## 12. INR trap with argatroban

### WHAT YOU SEE

The INR trap — argatroban falsely raises INR

### WHAT IT ENCODES

Argatroban prolongs the PT/INR more than other DTIs because it inhibits thrombin in the PT reagent, FALSELY elevating the INR. This is a major pitfall when transitioning a HIT patient from argatroban to warfarin: the combined INR overestimates the true warfarin effect, and stopping argatroban too early leaves the patient under-anticoagulated.

### BOARD TRAP

Practical board rule: during argatroban→warfarin overlap, target a HIGHER combined INR (often >4) per protocol, then STOP the argatroban infusion and RE-CHECK the INR ~2 hours later; only continue warfarin alone once the off-argatroban INR is therapeutic (2–3).

## 13. Both monitored by aPTT

### WHAT YOU SEE

The 'MONITORING' boxes under both fighters reading aPTT (target 1.5-3x baseline) — both DTIs are titrated to the aPTT, ACT in the cath lab.

### WHAT IT ENCODES

Both argatroban and bivalirudin are titrated to the aPTT (typically 1.5–2.5× baseline) for systemic anticoagulation; during PCI the activated clotting time (ACT) is used instead because of the much higher doses and shorter time frame.

### BOARD TRAP

Unlike heparin, DTIs require no antithrombin and are not reversed by protamine. Monitoring is aPTT/ACT — anti-Xa assays are for heparin/LMWH/fondaparinux, NOT for DTIs.